## Solutions

**Question 4.1**: Find the stationary point of the following functions and verify whether they are maxima or minima either by evaluating the function, or using the properties of the derivative.

(a) $f(x) = x^2 + x \implies \frac{df}{dx} = f'(x) = 2x + 1 \implies 2x + 1 = 0$
$x = -\frac{1}{2}$ is a stationary value and this is a minimum.

(b) $f(x) = 2x - x^2 \implies \frac{df}{dx} = f'(x) = 2 - 2x \implies 2 - 2x = 0$
$x = 1$ is a stationary value and this is a maximum.

(c) $f(x) = (x-2)^2 \implies \frac{df}{dx} = f'(x) = 2(x-2) \implies 2(x-2) = 0$
$x = 2$ is a stationary value and this is a minimum.

(d) $f(x) = (x-2)(3-x) \implies \frac{df}{dx} = f'(x) = (3-x) - (x-2) = 5 - 2x \implies 5 - 2x = 0$
$x = 2.5$ is a stationary value and this is a maximum.

(e) $f(x) = \frac{x^3}{3} - 4x^2 + 15x + 5 \implies \frac{df}{dx} = f'(x) = x^2 - 8x + 15 = (x-3)(x-5) \implies (x-3)(x-5) = 0$
$x = 3$ and $x = 5$ are stationary values, and they are a maximum and a minimum respectively.

(f) $f(x) = ln(x) - x \implies \frac{df}{dx} = f'(x) = \frac{1}{x} - 1 \implies \frac{1}{x} - 1 = 0$
$x = 1$ is a stationary value and this is a maximum.

**Question 4.2**: Write down an equation describing the problem: $x + y = 9$, where $x \times y^2$ is a maximum. Using the first equation we can write: $x = 9 - y$, sub this in the second to give:

$$f(y) = (9-y)y^2$$

so expanding, differentiating and setting to zero gives

$$f'(y) = 18y - 3y^2 = 0$$

Factorise to get

$$3y(6-y) = 0$$

which gives us $y = 0$ or $y = 6$ as stationary values. We see that $y = 0$ is a minimum and $y = 6$ is a maximum. Choosing $y = 6$ gives $x = 9 - 6 = 3$ and the maximum value is $3 \times 6^2 = 108$.

**Question 4.3**: We should write down the relevant equations and constraints. Since the box has a square base (i.e $l = w$) we can rewrite the equations as

$$V = lwh = w^2 h$$
$$10 = 2lw + 2wh + 2lh = 2w^2 + 4wh$$

Since we want to maximise $V$ and use a derivative, can we re-arrange the expressions to express $V$ with regards to $w$? We can. An expression for $h$ is given as:

$$h = \frac{10 - 2w^2}{4w} = \frac{5 - w^2}{2w}$$

Substitute this in to the expression for $V$ to arrive at the following expression that we will need to maximise:

$$V(w) = w^2 \left( \frac{5 - w^2}{2w} \right) = \frac{1}{2}(5w - w^3)$$

The derivative is given by:

$$\frac{dV}{dw} = \frac{1}{2}(5 - 3w^2)$$

Set this to 0 and rearrange to find we have one position solution as per the following:

$$w = \pm\sqrt{\frac{5}{3}} = \pm 1.2910$$

Since we cannot get negative width, we see the volume is maximised at

$$w = \sqrt{\frac{5}{3}} = 1.2910m$$

$$h = \frac{5 - w^2}{2w} = \frac{5 - \sqrt{\frac{5}{3}}^2}{2\sqrt{\frac{5}{3}}} = 1.2910m$$

so the optimum box is a cube. The maximum volume is:

$$V = w^2 h = \left( \sqrt{\frac{5}{3}} \right)^3 = 2.1517m^3$$

**Question 4.4**: This is very similar to the previous question. We should write down the relevant equations and constraints. The rectangular box has a square base (i.e $l = w$) and it is open so we can write the equations as

$$V = lwh = w^2 h$$

$$48 = lw + 2wh + 2lh = w^2 + 4wh$$

Note that we're leaving one square side off the constraint. Since we want to maximise $V$ and use a derivative, can we re-arrange the expressions to express $V$ with regards to $w$? We can. An expression for $h$ is given as:

$$h = \frac{48 - w^2}{4w} = \frac{48 - w^2}{4w}$$

Substitute this in to the expression for $V$ to arrive at the following expression that we will need to maximise:

$$V(w) = w^2\left(\frac{48-w^2}{4w}\right) = \frac{1}{4}(48w - w^3)$$

The derivative is given by:

$$\frac{dV}{dw} = \frac{1}{4}(48 - 3w^2)$$

Set this to 0 and rearrange to find we have one position solution as per the following:

$$w = \pm 4$$

Since we cannot get negative width, we see the volume is maximised at

$$w = 4m$$
$$h = \frac{48-w^2}{4w} = \frac{48-4^2}{4\times 4} = 2m$$

The maximum volume is:

$$V = w^2 h = 4\times 4\times 2 = 32m^3$$

**Question 4.5**:

(a) See table below: $r = 0.04$ m seems to minimise the surface area required.

| Radius | Height for volume to equal 330 ml | Surface area of aluminium |
|---|---|---|
| 0.010 | 1.050 | 0.067 |
| 0.020 | 0.263 | 0.036 |
| 0.030 | 0.117 | 0.028 |
| 0.040 | 0.066 | 0.027 |
| 0.050 | 0.042 | 0.029 |
| 0.060 | 0.029 | 0.034 |
| 0.070 | 0.021 | 0.040 |

(b) Since we need to minimise the surface area, we need to rearrange the volume to get an expression for $h$

$$h = \frac{330\times 10^{-6}}{r^2\pi}$$

Substitute this into the surface area equation to get a function in terms of $r$

$$A(r) = 2\pi r\left(\frac{330\times 10^{-6}}{r^2\pi}\right) + 2\pi r^2$$
$$= \frac{660\times 10^{-6}}{r} + 2\pi r^2$$

The derivative is given by:

$$\frac{dA}{dw} = -\frac{660 \times 10^{-6}}{r^2} + 4\pi r$$

Set this to 0 and rearrange to find we have one position solution as per the following:

$$\frac{660 \times 10^{-6}}{r^2} = 4\pi r \implies r = \left(\frac{660 \times 10^{-6}}{4\pi}\right)^{\frac{1}{3}}$$

we see the surface area is minimised at

$$w = r = \left(\frac{660 \times 10^{-6}}{4\pi}\right)^{\frac{1}{3}} = 0.0374$$
$$h = \frac{330 \times 10^{-6}}{r^2\pi} = 0.0749$$

**Question 4.6: A gold seam optimisation problem.**

(a) Firstly, write down the equation for profit, $P = G - C$:

$$P = 1 \times 10^5 - \frac{1 \times 10^5}{n+1} - 1000n$$

Find $\frac{dP}{dn}$:

$$\frac{dP}{dn} = -\frac{d}{dn}\left(\frac{1 \times 10^5}{n+1}\right) - 1000$$

Use the chain rule to differentiate $f = \frac{1\times10^5}{n+1}$ call $u = n+1$, so $f = \frac{1\times10^5}{u}$ and then

$$\frac{df}{dn} = \frac{df}{du} \times \frac{du}{dn} = -1 \times 10^5 u^{-2}$$

so we see that

$$\frac{dP}{dn} = \frac{1 \times 10^5}{(n+1)^2} - 1000$$

Now set to zero and solve for $n$ to find the stationary values:

$$\frac{1 \times 10^5}{(n+1)^2} - 1000 = 0$$
$$(n+1)^2 = 100$$
$$n^2 + 2n - 99 = 0$$
$$(n+11)(n-9) = 0$$

We see that this function has two stationary values:

$$n = -11, 9$$

So the answer is that the profit is maximised after 9 days. Substitute this into the formula for $P$ to get $P = 80,000$ profit after 9 days.

(b) To find the snag, we differentiate $G$ with respect to $n$:

$$G = 1\times 10^5\left[1-\left(0.01n^2+1\right)\exp\left(-0.1n\right)\right]$$
$$\frac{dG}{dn} = -1\times 10^5\times\frac{d}{dn}\left[\left(0.01n^2+1\right)\exp\left(-0.1n\right)\right]$$

Note that the first term is a constant so, when differentiated it goes to zero. To differentiate $\left(0.01n^2+1\right)\exp\left(-0.1n\right)$ we use the product rule. Split the function and differentiate:

$$u(n) = 0.01n^2+1, \quad \Longrightarrow \quad \frac{du}{dn} = 0.02n,$$
$$v(n) = \exp\left(-0.1n\right) \quad \Longrightarrow \quad \frac{dv}{dn} = -0.1\exp\left(-0.1n\right)$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dn} &= u\times\frac{dv}{dx}+v\times\frac{du}{dx}\\ &= -0.1\exp\left(-0.1n\right)\times\left(0.01n^2+1\right)+0.02n\times\exp\left(-0.1n\right)\\ &= -\exp\left(-0.1n\right)\left[-0.1\times\left(0.01n^2+1\right)-0.02n\right]\\ &= -\exp\left(-0.1n\right)\left[0.001n^2-0.02n+0.1\right]\end{aligned}$$

Now set to zero and solve for $n$ to find the stationary values:

$$\begin{aligned}-\exp\left(-0.1n\right)\left[0.001n^2-0.02n+0.1\right] &= 0\\ 0.001n^2-0.02n+0.1 &= 100\\ n^2-20n+100 &= 0\\ (n-10)^2 &= 0\end{aligned}$$

We see that this function has a stationary value:

$$n = 10$$

So the answer is that the snag occurs after 10 days. This is an example of a saddle point.

**Question 4.7: Geoengineering.** For the problem you should first find $\frac{dT}{dC}$,

$$T = 5000\exp\left(-dC\right)+C$$
$$\frac{dT}{dC} = -5000d\exp\left(-dC\right)+1$$

Now set to zero and solve for $C$ to find the stationary values:

$$\begin{aligned} -5000d\exp(-dC)+1 &= 0 \\ d\exp(-dC) &= \frac{1}{5000} \\ \exp(dC) &= 5000d \\ dC &= \ln(5000d) \\ C &= \frac{1}{d}\ln(5000d) \end{aligned}$$

Substitute $d = 2\times10^{-4}$ to find the amount to be spent on geoengineering, $C$, that minimises the total cost, $\chi$:

$$\begin{aligned} C &= \frac{1}{d}\ln(5000d) \\ C &= \frac{1}{2\times10^{-4}}\ln\left(5000\times2\times10^{-4}\right) \\ C &= 0 \end{aligned}$$

therefore this means that we can only *increase* the temperature by implementing the scheme! It should not go ahead.